M339D: April 6th, 2026.

More on Jensen's Inequality.

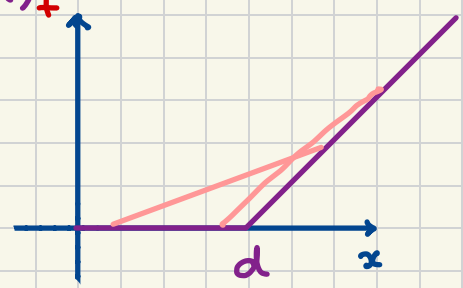
Example [cont'd].

iii. In classical insurance:

$$\begin{cases} X \ldots \text{ (ground-up) loss, i.e., severity r.v.} \\ d \ldots \text{ deductible} \end{cases}$$

The insurer pays

$$(X-d)_+, \text{ i.e., } g(x) = (x-d)_+$$



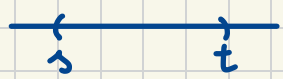
By Jensen's Inequality:

$$\mathbb{E}\left[(X-d)_+\right] \geq \left(\mathbb{E}[X]-d\right)_+$$

Log-Normal Stock Price.

Temporarily, fix a time horizon T.

$S(t)$, $t \in [0,T]$ .... time-t stock price



Recall:

$$R(s,t) := \ln\left(\frac{S(t)}{S(s)}\right) \iff S(t) = S(s)\, e^{R(s,t)}$$

In particular: $R(0,T)$ ... realized return for $[0,T]$

$$\boxed{\text{We model realized returns as normal.}}$$

$$R(0,T) \sim \text{Normal}\left(\text{mean} = m, \text{var} = \boxed{\nu^2}\right)$$

$\Rightarrow$ $S(T)$ is lognormal

$$\Rightarrow \mathbb{E}^*[S(T)] = \mathbb{E}\left[S(0)\, e^{R(0,T)}\right] = \boxed{S(0)\, e^{m + \frac{\nu^2}{2}}} \quad \star$$

Market model:

- **Riskless asset** w/ ccrfir r
- **Risky asset** : a non-dividend-paying stock w/ volatility σ

Q: Under the risk-neutral measure $\mathbb{P}^*$, we have

$$\boxed{\mathbb{E}^*[S(T)] = S(0)e^{rT}} \quad ☆☆$$

Equating ☆ and ☆☆, we get :

$$\boxed{m + \frac{\nu^2}{2} = rT} \checkmark$$

Q: $\mathrm{Var}[R(0,1)] = \cancel{?}\ \sigma^2$

$\Rightarrow \mathrm{Var}[R(0,T)] = \cancel{?}\ \boxed{\sigma^2 \cdot T = \nu^2}$

$$m = rT - \frac{\nu^2}{2} = rT - \frac{\sigma^2}{2}\cdot T = \left(r - \frac{\sigma^2}{2}\right)T$$

$$\Rightarrow \boxed{R(0,T) \sim \mathrm{Normal}\left(\text{mean} = \boxed{\left(r - \frac{\sigma^2}{2}\right)\cdot T},\ \text{var} = \sigma^2 \cdot T\right)}$$

Say, $Z \sim N(0,1)$.

Then, we can express $R(0,T)$ as

$$R(0,T) = \left(r - \frac{\sigma^2}{2}\right)\cdot T + \sigma\sqrt{T}\cdot Z$$

Then,

$$\boxed{S(T) = S(0)\,e^{R(0,T)} = S(0)\,e^{\left(r - \frac{\sigma^2}{2}\right)\cdot T + \sigma\sqrt{T}\cdot Z}}$$

Q: What is the median of $S(T)$ under the risk-neutral probability measure $\mathbb{P}^*$?

$\longrightarrow$: $\boxed{S(0)e^{\left(r - \frac{\sigma^2}{2}\right)\cdot T}}$

Note:

$$\frac{\text{mean of } S(T)}{\text{median of } S(T)} = \frac{\cancel{S(0)}\,\cancel{e^{rT}}}{\cancel{S(0)}\,e^{\left(\cancel{r} - \frac{\sigma^2}{2}\right)\cdot T}} = \boxed{e^{\frac{\sigma^2}{2}\cdot T}}$$

UNIVERSITY OF TEXAS AT AUSTIN

Problem Set # 13

Mean and median of the log-normal stock prices.

Problem 13.1. The current price of a non-dividend-paying stock is \$80 per share. Under the risk-neutral probability measure, its mean rate of return is 12% and its volatility is 30%.

Let $R(0,t)$ denote the realized return of this stock over the time period $[0,t]$ for any $t > 0$. Calculate $\mathbb{E}^*[R(0,2)]$.

$$\mathbb{E}^*[R(0,2)] = \left(r - \frac{\sigma^2}{2}\right)\cdot T = \left(0.12 - \frac{0.09}{2}\right)\cdot 2 =$$

$r = 0.12$

$$= (0.12 - 0.045)\cdot 2 = 0.15$$

Problem 13.2. A stock is valued at \$75.00. The continuously compounded, risk-free interest rate is 10.0% and the standard deviation of annualized returns is 25.0%. If the stock is lognormally distributed, what is the expected stock price after 2 years under the risk-neutral probability measure?

Problem 13.3. A non-dividend-paying stock is valued at \$55.00 per share. Its standard deviation of annualized returns is given to be 22.0%. The continuously compounded risk-free interest rate is 12%. If the stock price is modeled using the lognormal distribution (as discussed in class), what is the median of the stock price in 3 years under the risk-neutral probability measure?